KRITVA ROBOTICS

Robotic Actuator & Arm Mechanical Design Challenge

Intern Problem Statement • Critical Thinking • Proposed Draft Artwork

Assignment ID	Duration	Submission format
KR-MECH-INT-001	5 working days	CAD + calculations + concept artwork + short report

1. Problem Statement

Kritva is developing a humanoid robotics platform. Your assignment is to design a conceptual robotic elbow actuator and integrate it into a simplified robotic arm. The design should demonstrate basic actuator sizing, practical mechanical reasoning, component selection, and CAD modelling.

The goal is not to produce a manufacturing-ready design. We want to understand how you approach an open-ended engineering problem: how you turn requirements into calculations, make and document assumptions, compare options, recognize uncertainty, and communicate a design clearly.

2. Design Inputs

Parameter	Initial requirement
Payload mass	1.0 kg
Forearm and hand mass	1.5 kg
Payload centre-of-mass distance from elbow	350 mm
Forearm/hand centre-of-mass distance from elbow	180 mm
Target elbow output speed	45 RPM
Motor technology	BLDC/PMSM
Preferred transmission	Planetary gearbox; alternatives are allowed
Position feedback	Encoder
CAD tool	FreeCAD, SolidWorks, Fusion, or equivalent

Assume the forearm is horizontal for the initial static gravity calculation. If you use different assumptions for your proposed design, state them clearly. You may select reasonable link dimensions and commercial components, but support important choices with calculations or manufacturer data.

3. Assignment Tasks

A. Basic actuator sizing

- Calculate the elbow's static gravitational torque due to the payload and forearm/hand.
- Recommend a preliminary output torque target and explain the margin you choose.
- Estimate the motor speed and gearbox ratio needed to achieve the 45 RPM output target.
- Identify one commercially available motor and gearbox combination that appears suitable; include official datasheet or product references.

B. Mechanical concept and CAD

- Create a conceptual elbow actuator showing the motor, gearbox, output shaft, bearing support, housing, and forearm attachment.
- Integrate the elbow into a simplified arm with upper-arm link, forearm link, and wrist or payload interface.
- Show a plausible mechanical load path and consider access for assembly and maintenance.

- Include at least one assembly drawing or clearly labelled CAD view.

C. Design note

- Summarize your assumptions and design decisions.
- Provide a simple bill of materials (BOM).
- Identify two important design risks or challenges and suggest ways to address them.

4. Critical-Thinking Questions

Answer these questions briefly in your report. There may be more than one defensible answer; we are interested in your reasoning and how you handle trade-offs.

Q1 — Requirements conflict: Your selected motor and gearbox can produce enough torque, but the output speed is below 45 RPM. What would you investigate or change first, and why?

Q2 — Static versus dynamic loads: Your static calculation meets the target, but the elbow overheats during repeated movement. What could explain this, and what evidence would you collect?

Q3 — Load path: The gearbox manufacturer lists a suitable output torque, but the forearm applies a bending load to the output shaft. Why might the gearbox rating alone be insufficient? Sketch or describe your solution.

Q4 — Missing information: Which two additional requirements would you ask the project lead to clarify before freezing the design? Explain why they matter.

Q5 — Cost versus performance: A cheaper gearbox has noticeable backlash, while a more expensive option is stiffer and has lower backlash. Which would you choose for this concept and what information would affect your decision?

Q6 — Design review: Name one assumption in your design that could prove wrong. How would you test or validate it before committing to a prototype?

5. Proposed Draft Artwork to Submit

Alongside the CAD model, submit one page of proposed draft artwork that communicates your concept visually. This is an engineering concept illustration, not a polished marketing render. A neat hand sketch is acceptable if it is labelled and readable.

- One three-quarter/isometric view of the complete simplified robotic arm.
- One enlarged elbow-actuator view or exploded sketch showing motor, gearbox, shaft, bearings, housing, and encoder.
- Labels for the main components and arrows showing the approximate mechanical load path.
- Key dimensions: payload offset from elbow, forearm/hand centre of mass, and proposed link lengths.
- A small callout with the calculated static elbow torque and target output speed.
- A short note identifying one design trade-off or unresolved question.

Artwork tools are your choice: CAD screenshots, a hand sketch, PowerPoint, Inkscape, or another illustration tool. Do not spend time on branding or decorative effects; clarity and engineering communication matter more than visual polish.

6. Required Submission Package

Deliverable	Minimum expected content
1. Calculation sheet	Static torque, proposed torque target, motor speed/gear ratio reasoning
2. Component selection	One motor + gearbox proposal with datasheet/product references
3. CAD files	Elbow actuator concept and simplified robotic arm assembly
4. Drawing	One assembly drawing or labelled CAD view
5. Draft artwork	One-page concept illustration as described in Section 5
6. BOM	Main components, quantities, and manufacturer/model where known
7. Short report	Approximately 3–5 pages, including assumptions, critical-thinking answers, risks, and design rationale

7. Scope and Constraints

- A physical prototype is not required.
- Detailed finite-element analysis and manufacturing-ready drawings are not required.
- Do not purchase components or manufacture parts for this assignment.
- Use official datasheets or product references for selected components where possible.
- Clearly separate verified data from estimates and assumptions.
- If you find an inconsistency in the inputs, explain it and propose a resolution rather than ignoring it.

8. Final Review

Prepare a 10-minute presentation followed by a 10-minute technical discussion. Explain your torque calculation, component selection, mechanical load path, proposed artwork, major trade-offs, and what you would validate or improve next.

9. Evaluation Criteria (for candidate awareness)

Area	Weight
Engineering calculations and assumptions	25%
Motor/gearbox selection and evidence	20%
Mechanical design and CAD	25%
Critical thinking and trade-off reasoning	15%
Draft artwork and communication	10%
Documentation and organization	5%
Total	100%

Evaluation focuses on sound reasoning and a plausible concept, not on producing a production-ready actuator in five days. Candidates should identify uncertainties honestly and explain how they would resolve them.